

Financing Frictions: Theory

One wedge, six primitives

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Outline

General Setting

Micro-Foundations of the Wedge

One period, one input: a friction makes the firm invest too little

Static economy, one period, a representative firm

Technology in capital alone (labor held fixed):

$$y_i = z_i k_i^\alpha, \quad \alpha \in (0, 1)$$

- Why do we care?

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- Frictionless economy
 - All **NPV** > 0 projects are financed
 - Who is **rich** and who is **productive** does not matter

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- Why do we care?
- Frictionless economy
 - All **NPV** > 0 projects are financed
 - Who is **rich** and who is **productive** does not matter
- This lecture: **friction** makes firm i choose k_i **below** level a frictionless firm with same z_i

Frictionless benchmark: equal returns, net worth irrelevant

Entrepreneur with net worth a_i funds the shortfall $k_i - a_i$ at the common rate r :

$$\max_{k_i \geq 0} \underbrace{z_i k_i^\alpha}_{\text{revenue}} - \underbrace{r(k_i - a_i)}_{\text{external finance}} - \underbrace{r a_i}_{\text{own funds}} = z_i k_i^\alpha - r k_i$$

First-order condition and optimal scale:

$$\underbrace{\alpha z_i k_i^{\alpha-1}}_{\text{MRPK}_i} = r \quad \implies \quad k_i^* = \left(\frac{\alpha z_i}{r} \right)^{\frac{1}{1-\alpha}}$$

Net worth a_i cancels: every firm reaches k_i^* , so $\text{MRPK}_i = r$ for all i (Modigliani and Miller, 1958)

One friction, one wedge: $\text{MRPK}_i = r + \lambda_i$

Every core friction reopens the same gap in the benchmark relation:

$$\text{MRPK}_i = r + \lambda_i, \quad \lambda_i \geq 0$$

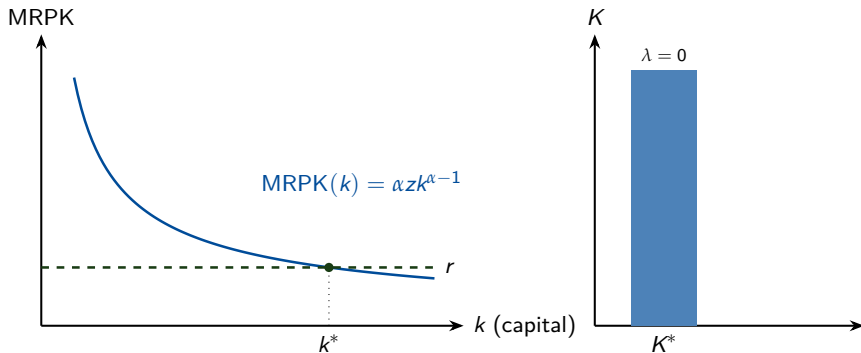
λ_i = shadow wedge that keeps firm i below its frictionless scale

– $\lambda_i > 0 \Rightarrow$ profitable marginal investment firm cannot fund at cost r

$\rightarrow$ Which benchmark assumptions must fail for $\lambda_i > 0$?

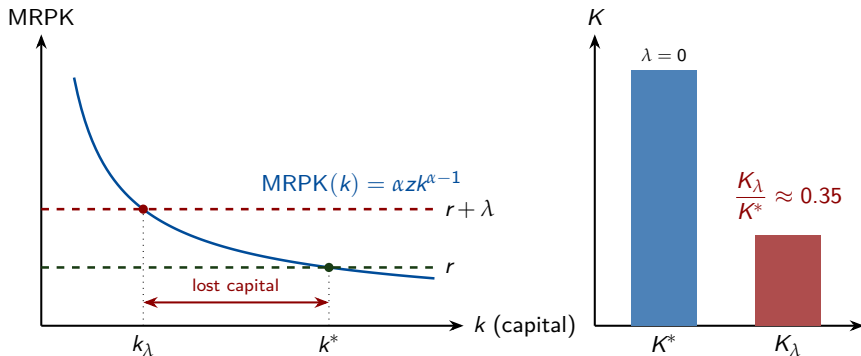
Why we care (today): a positive wedge sharply shrinks the capital stock

Frictionless firm invests to $MRPK = r$



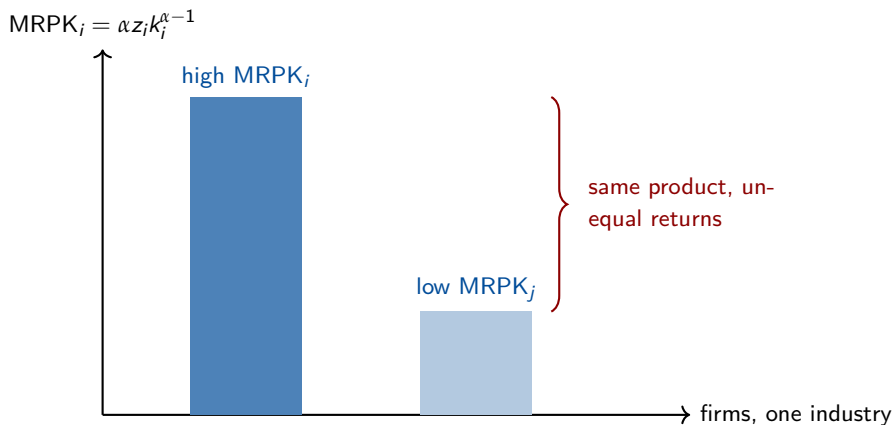
Why we care (today): a positive wedge sharply shrinks the capital stock

Frictionless firm invests to $\text{MRPK} = r$ Wedge firm stops at $\text{MRPK} = r + \lambda$



$\frac{K_\lambda}{K^*} = \left(\frac{r}{r + \lambda} \right)^{\frac{1}{1-\alpha}}$: doubling the cost of capital ($\lambda = r$, $\alpha = \frac{1}{3}$) leaves 35% of the stock cost

Why we care (misallocation): Different returns despite similar productivity



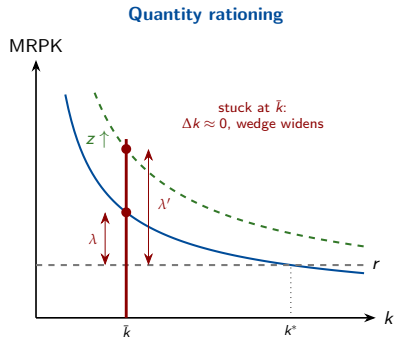
The wedge takes two forms: quantity vs price

- Quantity rationing: firm held **at a corner**, off its first-order condition
- Price wedge: firm stays interior, paying a higher marginal cost of funds

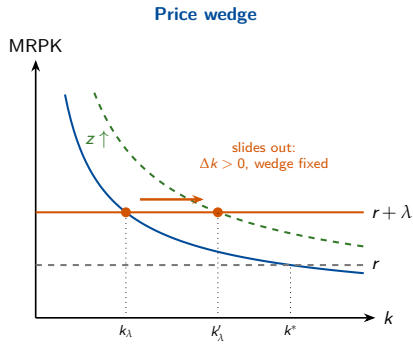
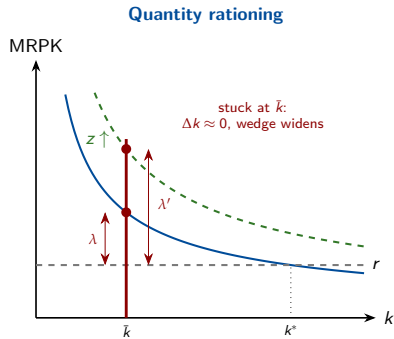
Takeaway 1

One underinvestment wedge, **two forms**: a quantity ceiling (firm at a corner) or a price premium (firm interior)

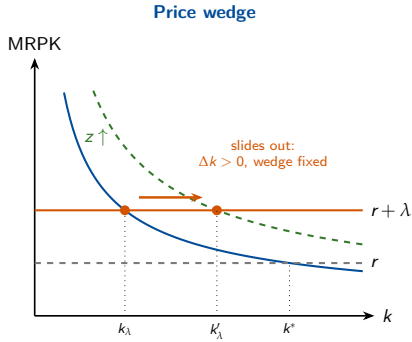
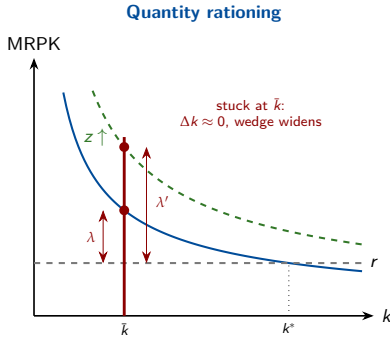
Same underinvestment, opposite reaction to a productivity shock



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Same underinvestment, opposite reaction to a productivity shock



💡 Takeaway 2

Same underinvestment, **opposite response to a shock**: holding net worth fixed, whether investment tracks productivity **identifies the regime** in the data

The two regimes hide in different data and call for different levers

Quantity rationing: wedge = shadow cost never paid in the loan rate

→ Two firms at the same rate can face different λ_i

Price wedge: firm actually pays $r + \lambda_i$, so the wedge shows up directly in the spread; identified by comparing exposed and unexposed borrowers

Takeaway 3

Read the wedge off the right data: MRPK; dispersion under rationing (never the loan rate), the spread under a price wedge

Credit mostly collateralized, and collateral does not (necessarily) track productivity

Loan = $f(\text{cash flow}, \text{collateral})$

- cash-flow-based lending is forward-looking; asset-based lending is backward-looking (Brooks and DAVIS, 2020)
- most credit is collateralized: Europe 70% (Degryse et al., 2025); Spain 40%, Peru 43% (Ivashina et al., 2022)

💡 Takeaway 4

Collateral, not productivity, gates credit: nothing makes the productive firm the one with collateral, so rationing tracks the productivity-wealth gap and misallocation arises

We derive limited enforcement next, the most transparent of the six generators of a collateral ceiling

Outline

General Setting

Micro-Foundations of the Wedge

Primitives open the wedge; ceilings and premia are their shadows

Primitive frictions: deep information, enforcement, or market-structure problems that block borrowing at the first-best rate and scale

Reduced-form borrowing technologies: the tractable objects those primitives generate, collateral ceilings, leverage caps, external finance premia

Six primitives, one wedge

#	Mechanism	What breaks	Side	Anchor model
1	Limited enforcement / collateral	Commitment	Borrower	Kiyotaki and Moore (1997)
2	Moral hazard / pledgeability	Hidden action	Borrower	Holmstrom and Tirole (1997)
3	Adverse selection / rationing	Hidden type	Borrower	Stiglitz and Weiss (1981)
4	Costly state verification	Hidden output	Borrower	Townsend (1979)
5	Intermediary constraints	Scarce bank capital	Lender	Gertler and Kiyotaki (2010)
6	Lender market power / hold-up	Imperfect competition	Lender	Rajan (1992)

Today we derive row 1 end to end; the other five share the wedge but break a different assumption

One model in full

Limited enforcement and collateral

Friction 1 breaks enforcement: the borrower can default and walk away

The benchmark assumed **perfect enforcement**; drop it, so courts enforce repayment only up to seizable collateral

Date-zero balance sheet, capital installed from own funds plus a loan:

$$k_i = a_i + b_i$$

If the entrepreneur defaults, the lender recovers only a fraction $\theta \in [0, 1]$ of installed capital:

$$\text{recovery on default} = \theta k_i$$

This θ is the **deep primitive** behind the reduced-form ceiling $b_i \leq \theta k_i$

Date 1: the loan is settled in cash at the gross rate $(1 + r) b_i$

Repayment incentive compatibility caps borrowing at θk_i

Repaying and continuing must beat defaulting and walking away:

$$\underbrace{z_i k_i^\alpha + k_i - (1+r)b_i}_{\text{repay, keep capital}} \geq \underbrace{z_i k_i^\alpha + (1-\theta)k_i}_{\text{default, abscond with } (1-\theta)k_i}$$

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Revenue $z_i k_i^\alpha$ cancels on both sides; the capital terms net to θk_i :

$$(1+r)b_i \leq \theta k_i$$

The ceiling grows with net worth and pledgeability

Substitute $b_i = k_i - a_i$ into $(1 + r)b_i \leq \theta k_i$:

$$(1 + r)(k_i - a_i) \leq \theta k_i \quad \implies \quad k_i \leq \bar{k}_i^{LE} \equiv \frac{(1 + r) a_i}{(1 + r) - \theta}$$

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- higher θ or higher a_i relaxes the ceiling ✓ higher r tightens it ✗

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- $\theta = 0$: **pure self-finance**, $\bar{k}_i^{LE} = a_i$

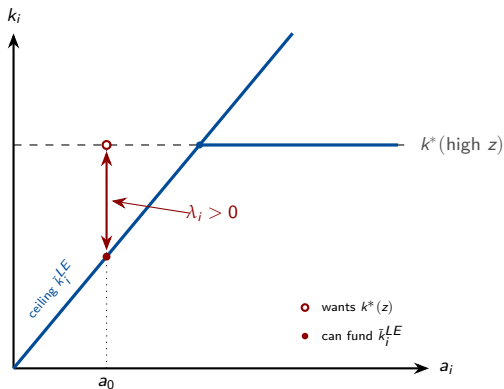
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- higher θ or higher a_i **relaxes** the ceiling ✓ higher r **tightens** it ✗
- $\theta = 0$: **pure self-finance**, $\bar{k}_i^{LE} = a_i$
- $\theta = 1$: **full pledgeability**, $\bar{k}_i^{LE} = \frac{(1+r)a_i}{r}$, **finite**, still not k^*
 - Low- a_i firm still **constrained** despite **full pledgeability** → why?

Constraint is the **gap** between the frictionless size and the observed size



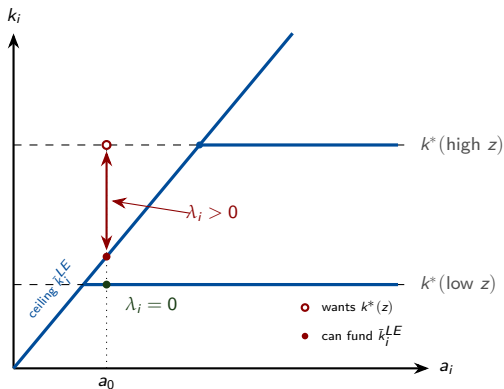
Firm invests $k_i = \min\{k_i^*, \bar{k}_i^{LE}\}$, binding whenever

$$a_i < \frac{(1+r)-\theta}{1+r} k_i^*(z_i)$$

so the threshold **rises with z_i** .

Fix net worth at a_0 . A **high- z** firm wants $k^*(\text{high } z)$, but the ceiling caps it: **constrained**, gap $= \lambda_i > 0$.

Constraint is the **gap** between the frictionless size and the observed size



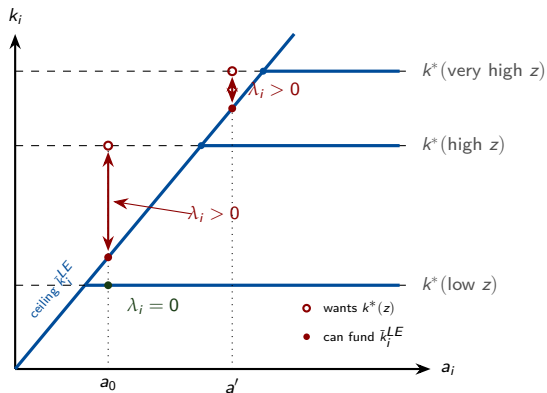
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Hold the *same* a_0 . A **low- z** firm wants less, the ceiling never binds: **unconstrained**, $\lambda_i = 0$, at identical net worth.

Constraint is the **gap** between the frictionless size and the observed size



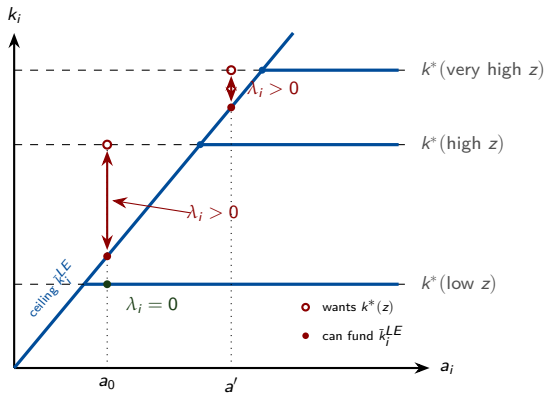
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so the threshold **rises** with z_i .

Now a richer firm at $a' > a_0$. If z is **very high**, the ceiling still caps it: **constrained** despite high net worth.

Constraint is the **gap** between the frictionless size and the observed size



Firm invests $k_i = \min\{k_i^*, \bar{k}_i^{LE}\}$, binding whenever

$$a_i < \frac{(1+r)^{-\theta}}{1+r} k_i^*(z_i)$$

so the threshold rises with z_j .

💡 Key takeaway

Low net worth firms can be unconstrained if their productivity is small

High net worth firms can be constrained if their productivity is very high

The shadow wedge λ_i^{LE} rises as net worth falls

For a constrained firm, plug $k_i = \bar{k}_i^{LE}$ into $MRPK_i = \alpha z_i k_i^{\alpha-1}$:

$$MRPK_i = \alpha z_i \left(\frac{(1+r) - \theta}{(1+r) a_i} \right)^{1-\alpha} > r$$

so the wedge is

$$\lambda_i^{LE} = \alpha z_i \left(\frac{(1+r) - \theta}{(1+r) a_i} \right)^{1-\alpha} - r$$

Larger when z_i is high, a_i is low, θ is low, the same directions that govern the ceiling

Constrained firms are young and asset-light

Most exposed: firms with **low net worth** relative to efficient scale, and firms whose assets are **hard to pledge**

- intangible-intensive and software firms
- human-capital-intensive and services firms
- young firms with little retained wealth

Least exposed: **asset-heavy** firms with high liquidation value, at the same productivity and net worth

Levers raise pledgeable value or net worth

Because the ceiling is $\bar{k}_i^{LE} = \frac{(1+r)a_i}{(1+r)-\theta}$, policy works through θ or a_i :

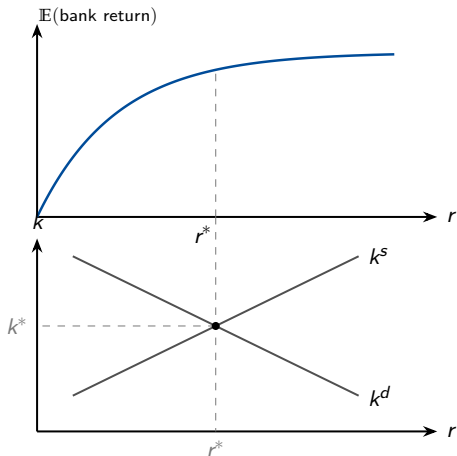
- stronger **creditor rights** and contract enforcement, raising θ
- **collateral and movable-asset registries**, making more assets seizable
- **public guarantees** that raise effective collateral value
- **wealth transfers** or start-up grants, raising a_i for high-ability, low-wealth entrants

Friction 3: Stiglitz-Weiss

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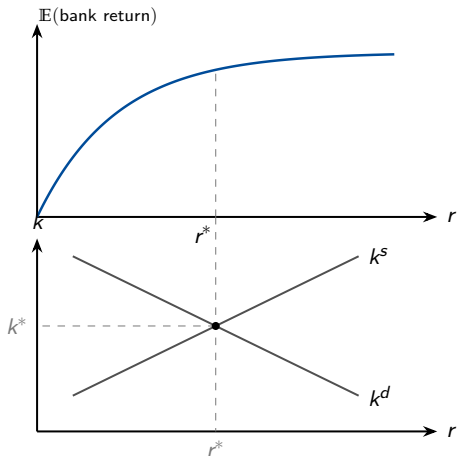
Stiglitz-Weiss in one slide: key intuition

Frictionless: $r^* = r^{\text{walras}}$

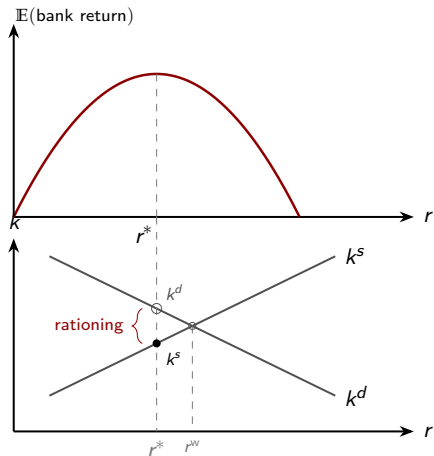


Stiglitz-Weiss in one slide: key intuition

Frictionless: $r^* = r^{\text{walras}}$



Friction: $r^* < r^{\text{walras}}$



Same wedge, a different lever for each primitive

The wedge λ_i is the **same**; the right **policy lever depends on which primitive** opened it:

- **Limited enforcement** → collateral registries, stronger creditor rights
- **Adverse selection** → information sharing, credit bureaus
- **Market power** → bank competition

Key challenge

The same λ_i is consistent with several primitives, so **identifying the exact friction in the data** is the hard part

Four takeaways from the models

💡 Takeaway 1

One wedge, **two forms**: a quantity ceiling (corner) or a price premium (interior)

💡 Takeaway 2

Different data reveal each regime: MRPK; dispersion under rationing, the spread under a price wedge

💡 Takeaway 3

Collateral, not productivity, gates credit, so rationing maps to the productivity-wealth gap and misallocation

💡 Takeaway 4

Same underinvestment, **opposite reaction to a shock**: rationed firms cannot act on good news, priced firms can

A static wedge is necessary, not sufficient, for misallocation

- benchmark: $\text{MRPK}_i = r$, net worth irrelevant
- friction: $\text{MRPK}_i = r + \lambda_i$, one reduced form, six primitives
- collateral: $\bar{k}_i^{LE} = \frac{(1+r)a_i}{(1+r)-\theta}$, wedge λ_i^{LE} rising as a_i falls
- quantity rationing strands firms at a corner; a price wedge keeps them interior

A static $\lambda_i > 0$ distorts investment today, but **self-financing can undo it**: persistent misallocation needs forces that keep recreating high- λ_i firms (Moll, 2014)

Thank you!

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